

Replication / Particle Physics

[Re] Do neutrinos dream in 5D? Partial replication of the Brane–Dirac spectrum and limits of open exclusion-contour reconstruction

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Edited by
(Editor)

Received
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Published
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DOI
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A partial replication of A. de Giorgi, D. Pasari, and J. Turner, “Do neutrinos dream in 5D? Towards a comprehensive extra-dimensional neutrino phenomenology,” JHEP 2026, 152 (2026).

Abstract We report a partial computational replication of the Brane–Dirac spectrum in Figure 1 of de Giorgi, Pasari, and Turner and examine the limits of extending that replication toward experiment-level exclusion contours. Starting from the published equations alone, we solve the infinite-tower eigenvalue equation for $m_D = 1$ and $\mu_1 = \{10, 1, 0.1\}$ and cross-check nine pre-specified test points with a mathematically distinct finite-KK calculation at $N = 5000, 10000, 20000$ followed by a leading- $1/N$ extrapolation; the largest mass difference is 1.29×10^{-9} , and the target paper’s 99% unitarity criterion is satisfied for all three parameter choices. All 3552 double-precision roots generated over the Figure 1 plotting range are independently resolved at 80-digit precision and agree within the Brent coordinate tolerance defined by $x_{\text{tol}} = 10^{-14}$ and $r_{\text{tol}} = 10^{-13}$, with a maximum normalized difference of 0.250331 and a maximum mass difference of 7.65×10^{-12} . A separate validation against the version-of-record Figure 1 finds graphical-footprint support for 12 of 15 pre-specified test points; two overlapping high-mode $\mu_1 = 10$ markers are supported by a control-calibrated occlusion assessment, while one $\mu_1 = 0.1$ point remains graphically unresolved. The pre-specified requirement of at least four supported points out of five for each μ_1 is satisfied. An exploratory open-data extension toward Figure 5 did not establish a unique pointwise correspondence with the published exclusion contour and is retained only to delimit the successful replication claim. The [Re] claim is therefore restricted to the theoretical Figure 1 subset; no exclusion-contour replication, new LED signal, or error in the original analysis is claimed.

1 Introduction

Large-extra-dimension (LED) neutrino models introduce Kaluza–Klein (KK) towers of gauge-singlet neutrino states that modify active-neutrino oscillations. Their phenomenology and experimental constraints have been studied for more than two decades^[1,2,3,4]. Recent analyses continue to refine reactor and accelerator constraints^[5,6].

The target of this replication is the 2026 analysis by de Giorgi, Pasari, and Turner^[5]. That article develops several five-dimensional neutrino-mass scenarios, derives their spectra and mixing patterns, and confronts them with oscillation data. Its Figure 1 presents the theoretical Brane–Dirac spectrum for $m_D = 1$ and $\mu_1 = \{10, 1, 0.1\}$; later figures present experiment-level exclusion contours. The published availability statement reports no deposited target-analysis code or associated analysis data.

Earlier work in this replication attempted to reconstruct the exclusion contours from public experimental products. Subsequent reanalysis showed

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The author has declared that no competing interests exist.
Code is available at <https://github.com/yokken0907/led-neutrino-open-reproduction-audit>.

that two initial interpretations were too strong: an apparent threshold-crossing difference had confounded KK truncation with unequal radius-grid resolution, and a binary Daya Bay “calibration failure” had compared our fixed-coordinate profiling implementation with a non-equivalent public full-fit construction. Both interpretations are withdrawn here.

The revised question is narrower: can the computational core of the published Brane–Dirac spectrum be independently reproduced from the equations in the target article, does that reconstruction agree with the actual published Figure 1 output, and how far can it be extended toward the experiment-level exclusion contours using open information? The answer reported here has two parts. Figure 1 is independently reproduced, numerically triangulated, and checked against the published graphical output. The Figure 5 extension is not established: selected digitized Daya Bay contour locations remain locally ambiguous in our independently constructed open-data comparator.

2 Replication target and claim boundary

2.1 Target result: Brane–Dirac spectrum

For the Brane–Dirac scenario, the target article writes the finite-tower eigenvalue condition and its infinite-tower closed form as^[5]

$$\sum_{n=0}^N \frac{(\chi_n m_D)^2}{m_\lambda^2 - \mu_n^2} = 1 \quad \xrightarrow{N \rightarrow \infty} \quad \pi \cot\left(\frac{\pi m_\lambda}{\mu_1}\right) = \frac{\mu_1 m_\lambda}{m_D^2}, \quad (1)$$

with $\mu_n = n\mu_1$, $\chi_0 = 1$, and $\chi_{n>0} = \sqrt{2}$. The corresponding infinite-tower normalization is

$$\mathcal{N}_\lambda = \left[\frac{1}{2} \left(\frac{\pi^2 m_D^2}{\mu_1^2} + \frac{m_\lambda^2}{m_D^2} + 1 \right) \right]^{-1/2}, \quad (2)$$

and unitarity implies

$$\sum_{\lambda} \mathcal{N}_\lambda^2 = 1. \quad (3)$$

The original Figure 1 plots Eq. (2) continuously and marks the physical roots of Eq. (1) for $m_D = 1$ and $\mu_1 = 10, 1, 0.1$.

2.2 What counts as replicated

The successful replication claim is limited to the computational content of Figure 1 and Eqs. (1)–(3). It does not include the Daya Bay or MINOS/MINOS+ likelihoods, the Figure 5 exclusion contours, or target-author implementation details that are not public. The Figure 5 work is retained because it tests the boundary of the open reconstruction. It is reported as an unresolved extension, not as a failed replication of the target article.

3 Independent Figure 1 implementation

3.1 Closed-form infinite-tower route

The primary implementation solves Eq. (1) directly in the dimensionless coordinate $y = m_\lambda/\mu_1$. Each positive physical root lies on the positive-cotangent

branch $n < y < n + 1/2$. Brent root finding was applied independently on each branch, and $\mathcal{N}_\lambda$ was evaluated from Eq. (2). No code from the target authors and no experimental likelihood implementation was imported into this calculation.

3.2 Finite-KK cross-check

A mathematically distinct cross-check evaluates the finite sum on the left-hand side of Eq. (1) directly with $\chi_0 = 1$, $\chi_{n>0} = \sqrt{2}$, and $\mu_n = n\mu_1$. Roots were obtained at $N = 5000, 10,000$, and $20,000$. The two finest values were combined with a leading- $1/N$ Richardson extrapolation. A leading $1/N$ correction is natural because, away from poles, the omitted large- n tail is governed by terms proportional to $1/n^2$, whose remainder after truncation at N is $O(1/N)$. Before inspecting the result, nine test points were pre-specified: $\mu_1 = 10, 1, 0.1$ crossed with mode labels $n = 0, 1, 5$. An absolute mass tolerance of 10^{-7} was used only as a numerical agreement criterion between the two routes; it is not a physics acceptance tolerance.

3.3 Unitarity and asymptotic diagnostics

For each published μ_1 , the number of roots included in Eq. (3) was increased until the partial sum exceeded 0.99. The 99% level is taken from the target article's numerical truncation criterion^[5]. The large-spacing and small-spacing limits discussed in the target article were evaluated as descriptive diagnostics; no additional pass/fail tolerance was introduced for those approximations.

4 Figure 1 replication results

4.1 Numerical triangulation

All nine pre-specified finite-sum/infinite-form comparisons pass. Table 1 reports the largest absolute mass difference within each published μ_1 group.

Table 1. Numerical triangulation of Figure 1 roots. Each row summarizes $n = 0, 1, 5$. The 10^{-7} acceptance level is a numerical agreement criterion between the two implementations, not a physics uncertainty.

μ_1/m_D	Maximum $ m_{\text{extrap}} - m_{\text{closed}} $	Status
10	2.31×10^{-11}	PASS
1	1.29×10^{-9}	PASS
0.1	1.31×10^{-10}	PASS

The first adaptive truncations to exceed the target 0.99 criterion were 64, 256, and 32768 modes for $\mu_1 = 10, 1, 0.1$, respectively, giving partial sums 0.9996850, 0.9921727, and 0.9938966. The independently generated spectrum is shown in Figure 1. The lightest-state overlap decreases from $\mathcal{N}_\lambda = 0.9837$ at $\mu_1 = 10$ to 0.4249 at $\mu_1 = 1$ and 0.0450 at $\mu_1 = 0.1$.

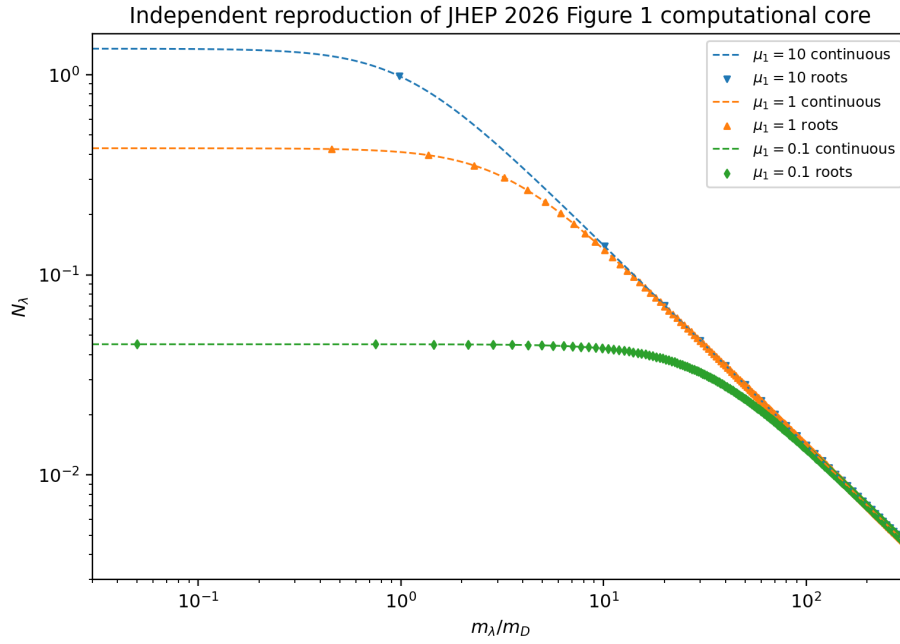


Figure 1. Independent reproduction of the computational core of the target article’s Figure 1. Dashed curves evaluate the published infinite-tower normalization continuously; markers show independently solved physical roots for $m_D = 1$ and $\mu_1 = 10, 1, 0.1$. This panel is generated from the published equations and is not copied from the target article.

4.2 Direct high-precision validation of all computed roots in the plotting range

The first implementation included an additional raw equation-residual acceptance criterion, $|F(y)| < 10^{-9}$, for every plotted root of

$$F(y) = \pi \cot(\pi y) - \left(\frac{\mu_1}{m_D} \right)^2 y. \quad (4)$$

That automatic criterion failed because the maximum raw residual was 2.67×10^{-6} . This result is retained unchanged in the computational record. Near cotangent poles, however, a raw function residual is a poorly conditioned proxy for root-coordinate accuracy.

To avoid relying on a derivative-scaled estimate as the final accuracy statement, all 3552 double-precision roots generated over the Figure 1 plotting range were independently re-solved with 80 decimal digits using the same published equation but a high-precision numerical implementation. The direct coordinate difference $|y_{\text{double}} - y_{\text{hp}}|$ was compared with the Brent coordinate tolerance $x_{\text{tol}} + r_{\text{tol}}|y|$, with $x_{\text{tol}} = 10^{-14}$ and $r_{\text{tol}} = 10^{-13}$. All 3552 roots lie within that tolerance. The maximum difference-to-tolerance ratio is 0.250331, the median is 0.002875, and the 99th percentile is 0.239409. The largest absolute mass difference over all roots is 7.65×10^{-12} . Thus the earlier raw-residual failure is a conditioning-sensitive diagnostic failure, while the computed root coordinates are directly confirmed at high precision. The plotting routine thins the densely spaced $\mu_1 = 0.1$ marker series for readability; the count 3552 refers to the complete computed root set over the plotting range, not to the number of marker glyphs visibly drawn.

4.3 Comparison with the published Figure 1 output

A separate graphical validation tests whether the reconstructed roots are consistent with the actual output shown in the target article rather than only with internally consistent calculations. Fifteen test points were pre-specified, five for each published μ_1 . The version-of-record Figure 1 was rendered at 400, 600, and 800 dpi and analyzed with three color-saturation settings. Axis calibration was required to be consistent across renderings; the maximum fractional cross-DPI deviations were 2.11×10^{-4} in x and 7.06×10^{-4} in y . We use *graphical-footprint support* when the theoretical point falls within the detected colored marker footprint expanded by one raster pixel and the measured axis-calibration residual. This criterion tests consistency with the rasterized marker footprint; it is not a precision-coordinate agreement criterion. For densely spaced low- μ_1 modes, several rasterized markers can merge into one connected colored component, so footprint support does not identify the centroid of an individual marker.

Twelve of the 15 test points have direct graphical-footprint support. The initial footprint analysis classified the $\mu_1 = 10$, $n = 5$ and $n = 10$ points as mismatches because blue or green series overplot part of the purple downward-triangle marker. Their original automated classifications are retained. A separate occlusion assessment used the fully visible $n = 0$ marker as the glyph reference and $n = 1, 2$ as registration controls. The largest control registration residual was 1.26247 pixels on a 400-dpi-equivalent scale; adding exactly one 400-dpi raster pixel set the target registration threshold to 2.26247 pixels before the two targets were evaluated. Both controls pass registration in all nine dpi/saturation combinations. The $n = 2$ control additionally exhibits the prescribed overplot/occlusion signature in all nine combinations, whereas $n = 1$ does not. That signature requires the visible purple component to be at most 80% of the reference width and height, blue/green pixels to occupy at least 10% of the predicted full-marker box, and the visible purple bounding box to remain geometrically contained within the reference-derived full-marker box with a one-scaled-pixel allowance. The two target points $n = 5$ and $n = 10$ pass both registration and the occlusion-signature criterion in all nine combinations, with maximum registration residuals of 1.18605 and 1.63669 pixels, respectively. One test point, $\mu_1 = 0.1$, $n = 20$, remains insufficiently supported graphically.

Table 2. Published-output validation at the 15 pre-specified Figure 1 test points. “Footprint” means consistency with the detected rasterized marker footprint; “Occlusion” means support from the separately calibrated marker-overlap assessment; “Unresolved” means insufficient graphical support. The pre-specified adequacy rule is at least four supported points out of five for each μ_1 .

μ_1	Footprint	Occlusion	Unresolved	Supported
10	3	2	0	5/5
1	5	0	0	5/5
0.1	4	0	1	4/5

Figure 2 shows the target panel with the replica test points overlaid in the same calibrated coordinate system. The graphical validation is deliberately bounded: it verifies consistency with the published rasterized output and does not reconstruct the target authors’ code or likelihood infrastructure.

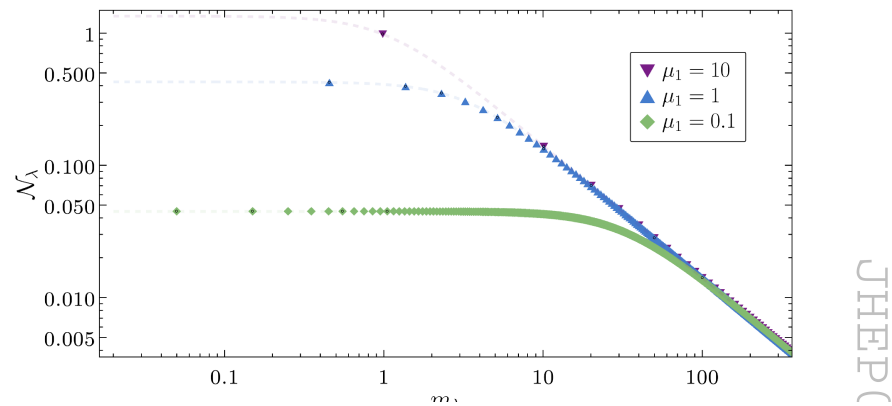


Figure 2. Published-output comparison. Replica test-point locations are overlaid on the published Figure 1 panel. The underlying target panel is adapted from de Giorgi, Pasari, and Turner^[5], distributed under CC BY 4.0; the overlay and annotations are from the present replication. The two high-mode $\mu_1 = 10$ points require the separate occlusion assessment described in the text, while $\mu_1 = 0.1, n = 20$ remains graphically unresolved.

5 Limits of the Figure 5 extension

The target article’s Figure 5 gives experiment-level exclusion contours for the Dirac-brane scenario^[5]. We explored whether representative locations on the published Daya Bay contour could be mapped pointwise to an independently constructed comparator built from open experimental products. This extension does not replay the target authors’ likelihood construction, which is not publicly available in sufficient detail for an exact implementation-level reproduction.

The exploratory comparison did not establish a unique pointwise correspondence: within the digitization envelopes of the selected locations, the open comparator exhibited multiple local threshold crossings. We therefore treat the Figure 5 exercise only as a scope-limiting diagnostic. Its detailed numerical scan values are not part of the present [Re] result and are not used as evidence for successful replication.

A separate reanalysis also showed that an earlier apparent dependence of fine threshold-crossing structure on KK truncation had been confounded by unequal radius-grid sampling. That interpretation is withdrawn. These observations do not imply that the published Figure 5 contours are wrong or non-replicable; they show only that the open reconstruction tested here does not provide a unique pointwise replay of them.

6 Auxiliary control and withdrawn calibration interpretation

An earlier auxiliary Daya Bay control used the public data release^[7] to compare our fixed-coordinate profiling implementation with the public covariance-fit route. The public route reproduced its own stored reference result in the controlled environment, whereas the fixed-coordinate implementation

Table 3. All 15 pre-specified graphical test points. Raster medians summarize successful render/detection variants and should not be interpreted as precision coordinate estimates. “Footprint” and “Occlusion” have the meanings defined in Table 2.

μ_1	n	Theory ($m_\lambda, \mathcal{N}_\lambda$)	Raster median ($m_\lambda, \mathcal{N}_\lambda$)	Initial	Final support
10	0	(0.98385, 0.98374)	(0.98582, 1.00491)	support	Footprint
10	1	(10.09899, 0.13929)	(10.13398, 0.14371)	support	Footprint
10	2	(20.04987, 0.07044)	(20.53426, 0.07345)	support	Footprint
10	5	(50.01999, 0.02827)	(51.72049, 0.02962)	mismatch	Occlusion
10	10	(100.01000, 0.01414)	(103.81999, 0.01482)	mismatch	Occlusion
1	0	(0.45429, 0.42494)	(0.45526, 0.41473)	support	Footprint
1	1	(1.36917, 0.39615)	(1.37174, 0.38633)	support	Footprint
1	2	(2.29891, 0.35186)	(2.30721, 0.34346)	support	Footprint
1	5	(5.17371, 0.23052)	(5.19027, 0.22560)	support	Footprint
1	10	(10.09603, 0.13316)	(10.13788, 0.13030)	support	Footprint
0.1	0	(0.04995, 0.04499)	(0.04999, 0.04488)	support	Footprint
0.1	1	(0.14985, 0.04499)	(0.15013, 0.04489)	support	Footprint
0.1	5	(0.54944, 0.04499)	(0.55059, 0.04489)	support	Footprint
0.1	10	(1.04894, 0.04497)	(1.15080, 0.04488)	support	Footprint
0.1	20	(2.04793, 0.04490)	—	unresolved	Unresolved

returned a different optimum because the two procedures were not computationally equivalent. We therefore withdraw the earlier interpretation of this discrepancy as a failure of the public Daya Bay model or data release. This auxiliary control is retained only to explain the scope correction that led to the present Figure 1-only replication claim. It is not part of the manuscript-facing executable [Re] workflow, and no quantitative auxiliary fit result is used as evidence for the successful replication reported here.

7 Reproducibility status

The public repository provides one current entry point for the Figure 1 replication. From a fresh checkout, run

```
bash reproduce_figure1_revision2.sh
```

This command runs the published-equation spectrum calculation, the finite-KK cross-check, the direct 80-digit comparison of all 3552 roots generated over the Figure 1 plotting range, and the full graphical validation from the version-of-record PDF. The graphical step verifies the PDF SHA-256 before rasterization, repeats the 400/600/800-dpi axis calibration and marker-footprint analysis, generates the overlay and 15-point table, and then applies the separately calibrated occlusion assessment to the two overlapping $\mu_1 = 10$ markers. The Dockerfile invokes the same current entry point.

The repository commit used for this manuscript is 22b24bf63634. GitHub Actions run 36585778666 executed the complete workflow from a fresh hosted checkout and completed successfully. The archived result artifact has ID 11041541753 and SHA-256 digest 547010bed5bd33ce...; the complete digest is recorded in the repository evidence record. The initial raw-residual failure and the initial two graphical-footprint mismatches remain preserved in the generated outputs rather than being overwritten by the subsequent high-precision and occlusion checks.

8 Limitations and claim boundaries

The successful result remains deliberately narrow. This work establishes an independent reproduction of the Figure 1 Brane-Dirac computational spectrum from published equations and demonstrates consistency with the pub-

lished graphical output under a pre-specified, multi-resolution validation. It does not reconstruct the target authors' full numerical infrastructure, experimental likelihoods, or Figure 5 exclusion contours.

The graphical validation is not equivalent to access to underlying target numerical arrays. Twelve of 15 test points are supported directly by their rasterized graphical footprints, two are supported only through the separately calibrated occlusion assessment, and one remains graphically unresolved. The pre-specified per- μ_1 adequacy rule is satisfied, but the manuscript does not claim that all 15 published points were independently recovered as exact numerical coordinates.

Likewise, the Figure 5 comparator uses open experimental products and a related public implementation, not the target authors' unavailable likelihood pipeline. Multiple local threshold crossings diagnose that open reconstruction. The target paper's experiment-level results are neither validated nor falsified here.

9 Conclusion

The Brane-Dirac spectrum underlying Figure 1 of de Giorgi, Pasari, and Turner is independently reproduced from the published equations. A closed-form infinite-tower implementation and a mathematically distinct finite-KK cross-check agree at nine pre-specified test points to at worst 1.29×10^{-9} in mass, and the target article's $> 99\%$ unitarity criterion is satisfied for all three published μ_1 choices. Direct 80-digit re-solution of all 3552 roots generated over the Figure 1 plotting range confirms every double-precision root within the Brent solver-coordinate tolerance $x_{\text{tol}} = 10^{-14}$, $r_{\text{tol}} = 10^{-13}$.

A separate published-output validation connects that computational agreement to the actual Figure 1: 12 test points are supported directly by their rasterized marker footprints, two high-mode points are supported by a control-calibrated marker-occlusion assessment, and one test point remains graphically unresolved while the pre-specified adequacy criterion is still satisfied. The successful claim therefore extends beyond internal numerical consistency to a scope-limited comparison with the published output, without claiming access to the target code or numerical arrays.

The replication does not extend to the tested Figure 5 exclusion locations: the exploratory open-data comparator did not establish a unique pointwise mapping to the published contour. An auxiliary Daya Bay control likewise served only to correct an earlier scope interpretation by showing that our fixed-coordinate profiling implementation was not computationally equivalent to the public reference route. Neither auxiliary analysis is part of the successful [Re] claim.

Replication-status statement

The [Re] prefix is used only for the independently reproduced Figure 1 Brane-Dirac spectrum subset. The experiment-level Figure 5 exclusion contours remain unresolved in the open reconstruction reported here and are explicitly excluded from the successful-replication claim. We do not use [¬Re] because the present study does not establish that the original exclusion analysis is non-replicable or mistaken.

AI-use disclosure

OpenAI ChatGPT was used as an assistive tool for code drafting, validation organization, manuscript drafting, and consistency checking. Numerical claims in this article were accepted only when reproduced by executable scripts and traced to versioned computational outputs or primary sources. The author selected the research questions, executed the computational packages, reviewed the outputs, and remains responsible for the scientific claims and the manuscript.

Competing interests

The author declares no competing interests.

Code and data

Public repository: <https://github.com/yokken0907/led-neutrino-open-reproduction-audit>. The repository commit used for this manuscript is 22b24bf63634, exercised by GitHub Actions run 36585778666. The one-command entry point is the shell runner stated in Section 7. Versioned computational evidence and cryptographic hashes are retained in the repository and accompanying evidence package. The Daya Bay public release discussed only in the auxiliary historical section is cited as a literature/data reference and is not an input to the successful Figure 1 replication.

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